Ain Shams University
Faculty of Engineering
CSE343 - Web Development, Fall 2026

Lab 4 Report: Todo App (TypeScript)

Name: Rodina Mohamed

Student ID: 23P0191

GitHub: https://github.com/Rodina761/webDevelopments-lab-submissions-

The todo app is written in TypeScript (script.ts) and compiled to JavaScript with `npx tsc`. The HTML page loads the generated script.js.

Features

- Add a todo with the Add button or by pressing Enter
- Mark a todo as done with the checkbox, delete it with the trash button
- Double-click a todo to edit it, press Enter to save
- Filter by All, Active or Completed
- Items left counter, Clear completed and Clear all

Each todo is an object of a Todo class (id, text, done) and all todos are stored in a typed array. The list is rebuilt by a render function after every change. filter() is used for the filters, deleting and clearing, and the spread operator is used when adding.

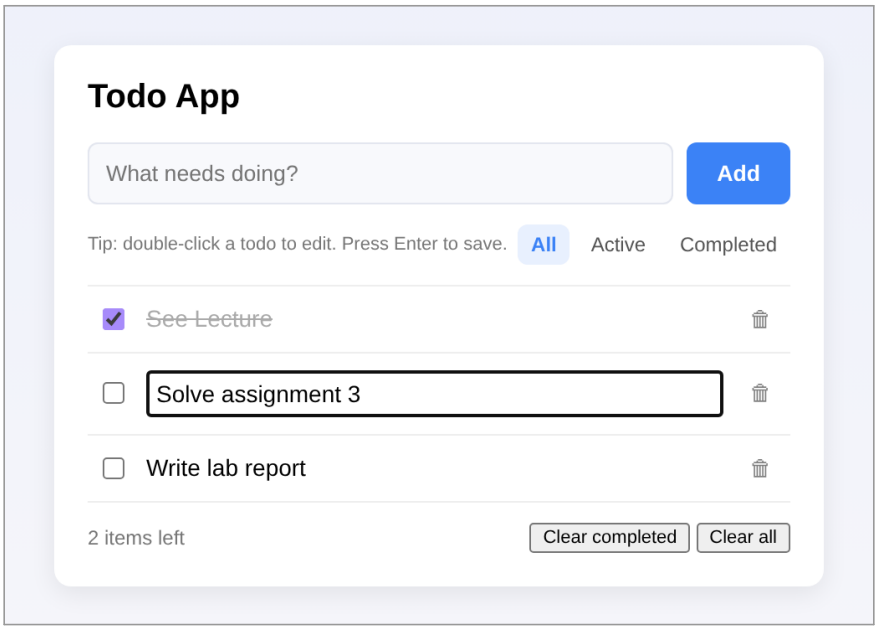


Figure 1: One todo completed and one being edited

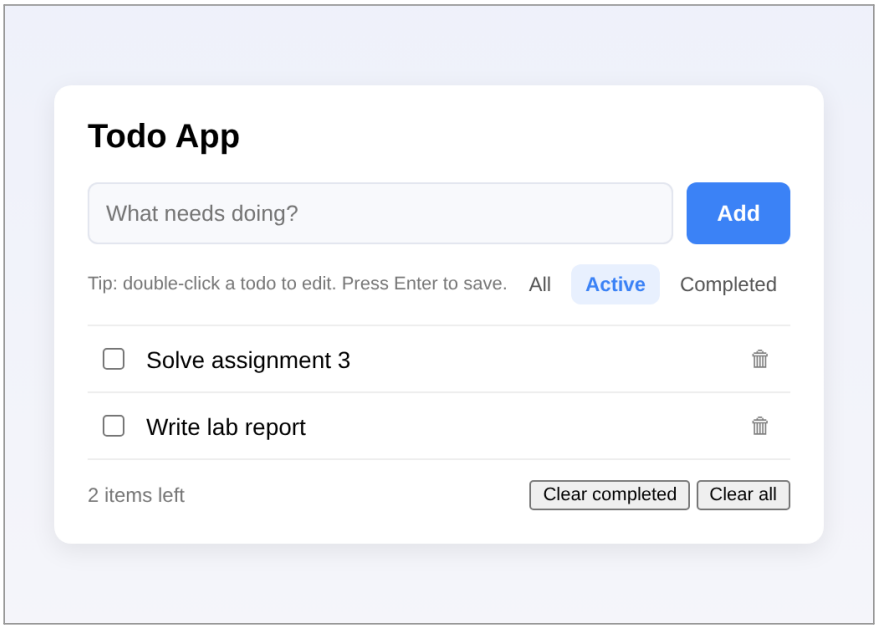


Figure 2: Active filter, only unfinished todos are shown

Source files

- index.html: page structure
- style.css: card layout and styling
- script.ts: TypeScript source code
- script.js: JavaScript generated by tsc
- tsconfig.json, package.json: TypeScript project setup